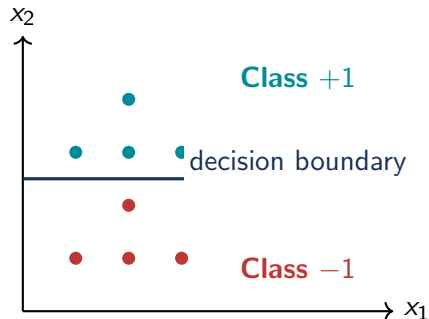


Support Vector Machines

3. From Separation to Maximum Margin

From Separation to Maximum Margin

Suppose we have two classes of observations.



The problem

This line separates the two classes perfectly.

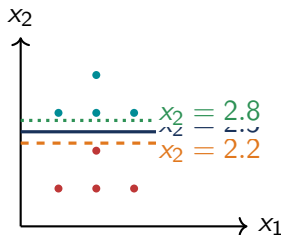
But **why should we choose this line rather than another separating line?**

Several Separating Boundaries Can Exist

Use the **same dataset**:

+1	-1
(1, 3), (2, 4), (3, 3), (2, 3)	(1, 1), (2, 2), (3, 1), (2, 1)

Three different horizontal lines all separate the classes:

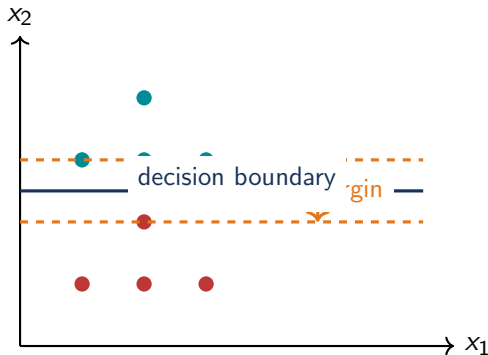


All three boundaries classify the displayed training observations correctly. Therefore:

Training accuracy alone does not determine the best boundary.

What Is a Margin?

The **margin** measures how far the closest observations are from the decision boundary.



Intuition

A larger margin means the classifier has a larger “buffer zone” between the two classes.

Functional Margin

For a point $(\mathbf{x}_i, y_i)$, define the classifier score

$$f(\mathbf{x}_i) = \mathbf{w}^\top \mathbf{x}_i + b.$$

The **functional margin** is

$$\hat{\gamma}_i = y_i(\mathbf{w}^\top \mathbf{x}_i + b)$$

where

$$y_i \in \{-1, +1\}.$$

Small numerical example

Let $\mathbf{w} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, $b = -5$. For the positive point

$$\mathbf{x}_i = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad y_i = +1,$$

$$\hat{\gamma}_i = (+1)[0(2) + 2(3) - 5] = 1.$$

Geometric Margin

The functional margin depends on the scale of $\mathbf{w}$.

To obtain an actual geometric quantity, divide by $\|\mathbf{w}\|$:

$$\gamma_i = \frac{y_i(\mathbf{w}^\top \mathbf{x}_i + b)}{\|\mathbf{w}\|}$$

This is the **geometric margin**.

Numerical example

For $\mathbf{w} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, $b = -5$, we have $\|\mathbf{w}\| = \sqrt{0^2 + 2^2} = 2$.

For $\mathbf{x} = (2, 3)$ and $y = +1$:

$$\hat{\gamma} = 1.$$

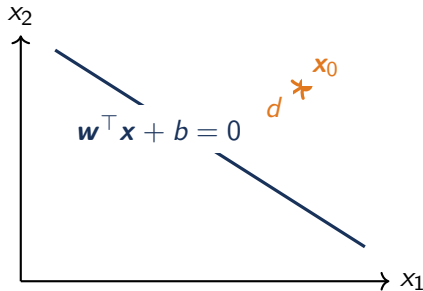
Therefore

$$\gamma = \frac{1}{2} = \boxed{0.5}.$$

Distance of a Point from a Hyperplane

For the hyperplane $\mathbf{w}^\top \mathbf{x} + b = 0$, the perpendicular distance from a point $\mathbf{x}_0$ is

$$d = \frac{|\mathbf{w}^\top \mathbf{x}_0 + b|}{\|\mathbf{w}\|}$$



Geometric meaning

The distance is measured along the direction **perpendicular** to the hyperplane.

Numerical Example: Point-to-Boundary Distance

Consider the decision boundary $2x_2 - 5 = 0$. Therefore, $\mathbf{w} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, $b = -5$.

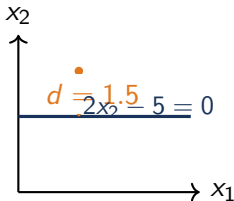
Take the point

$$\mathbf{x}_0 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Step 1: Numerator $|\mathbf{w}^\top \mathbf{x}_0 + b| = |0(2) + 2(4) - 5| = |3| = 3$.

Step 2: Norm $\|\mathbf{w}\| = 2$.

Step 3: Distance $d = \frac{3}{2} = 1.5$



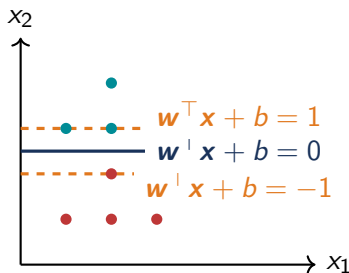
Positive and Negative Margin Boundaries

The decision boundary is $\mathbf{w}^\top \mathbf{x} + b = 0$.

SVM uses the two reference boundaries $\mathbf{w}^\top \mathbf{x} + b = 1$ and $\mathbf{w}^\top \mathbf{x} + b = -1$. For our example,

$$\mathbf{w} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad b = -5.$$

Therefore: $2x_2 - 5 = 1 \Rightarrow x_2 = 3$ and $2x_2 - 5 = -1 \Rightarrow x_2 = 2$.



Why Do We Use $+1$ and -1 ?

The equations

$$\mathbf{w}^\top \mathbf{x} + b = 1$$

and

$$\mathbf{w}^\top \mathbf{x} + b = -1$$

are a convenient **scaling convention**.

For the closest observations, SVM chooses the scale so that

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1.$$

Thus:

$$y_i = +1 \Rightarrow \mathbf{w}^\top \mathbf{x}_i + b = +1,$$

$$y_i = -1 \Rightarrow \mathbf{w}^\top \mathbf{x}_i + b = -1.$$

$$(2, 3), \quad y = +1 :$$

$$2(3) - 5 = 1.$$

And

$$(2, 2), \quad y = -1 :$$

$$2(2) - 5 = -1.$$

Important

The observations satisfying

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$$

will later become the key observations called **support vectors**.

Deriving the Margin Width

Distance from

$$\mathbf{w}^\top \mathbf{x} + b = 0$$

to

$$\mathbf{w}^\top \mathbf{x} + b = 1$$

is

$$d_+ = \frac{|1 - 0|}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}.$$

Similarly,

$$d_- = \frac{|0 - (-1)|}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}.$$

Therefore the total width is

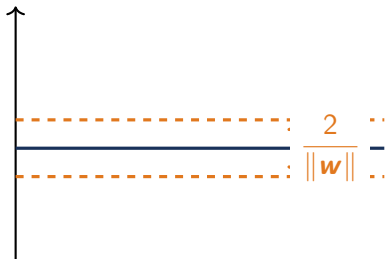
$$M = d_+ + d_-.$$

Hence

$$M = \frac{1}{\|\mathbf{w}\|} + \frac{1}{\|\mathbf{w}\|}$$

and finally

$$M = \frac{2}{\|\mathbf{w}\|}$$



Numerical Example: Margin Width

For our classifier,

$$\mathbf{w} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

Therefore,

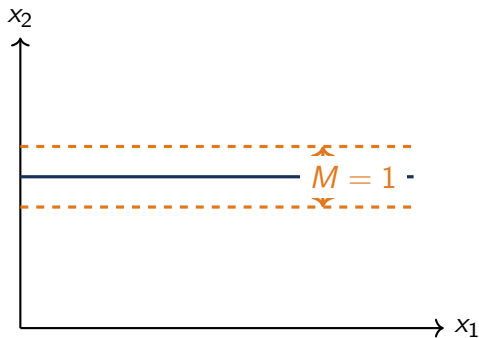
$$\|\mathbf{w}\| = 2.$$

The total margin width is

$$M = \frac{2}{\|\mathbf{w}\|} = \frac{2}{2} = \boxed{1}.$$

The distance from the decision boundary to either margin boundary is

$$\frac{1}{\|\mathbf{w}\|} = \frac{1}{2} = \boxed{0.5}.$$



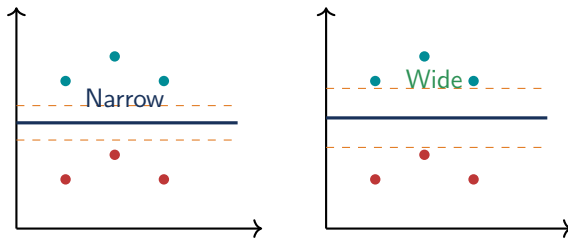
Key relationship

Large $\|\mathbf{w}\| \Rightarrow$ small margin

Small $\|\mathbf{w}\| \Rightarrow$ large margin

Why Is a Large Margin Desirable?

Consider two possible boundaries.



Why prefer a wide margin?

A wide margin gives the classifier more separation from the training observations. This generally makes the decision boundary more **robust to small changes in future observations**.

Maximum-Margin Principle

We have established

$$M = \frac{2}{\|\mathbf{w}\|}$$

Therefore, maximizing the margin is equivalent to minimizing

$$\|\mathbf{w}\|.$$

For mathematical convenience, SVM minimizes

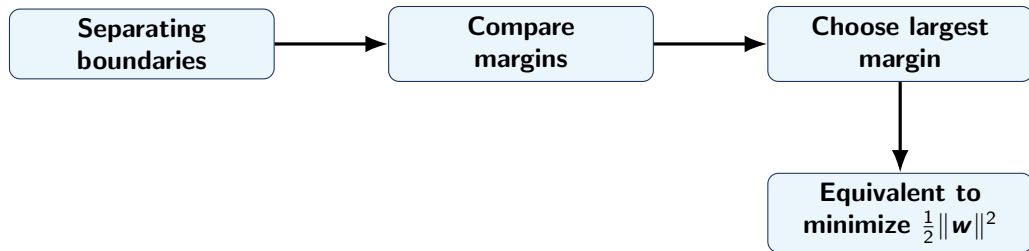
$$\frac{1}{2} \|\mathbf{w}\|^2$$

instead.

Thus the basic idea is

$$\text{Maximum margin} \iff \text{Minimum } \frac{1}{2} \|\mathbf{w}\|^2$$

while still requiring correct classification.



Small Numerical Example: Which Boundary Is Better?

Suppose two candidate separating classifiers have

$$\mathbf{w}_A = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \quad \mathbf{w}_B = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

For A:

$$\|\mathbf{w}_A\| = 4, \quad M_A = \frac{2}{4} = 0.5.$$

For B:

$$\|\mathbf{w}_B\| = 2, \quad M_B = \frac{2}{2} = 1.$$

Therefore,

$$M_B > M_A$$

and the maximum-margin principle prefers B.

Candidate A

$$\|\mathbf{w}_A\| = 4, \quad M_A = 0.5$$

Candidate B

$$\|\mathbf{w}_B\| = 2, \quad M_B = 1$$

SVM preference

Larger margin $\Rightarrow$ preferred boundary

From Geometry to the SVM Optimization Problem

We now know:

$$M = \frac{2}{\|\mathbf{w}\|}.$$

So SVM wants a large margin.

Equivalently:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2.$$

But we cannot minimize $\|\mathbf{w}\|$ without restriction.

The observations must remain correctly classified.

For labels

$$y_i \in \{-1, +1\},$$

the basic constraint will be

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$$

for every training observation.

Therefore the hard-margin SVM problem is

$$\begin{array}{ll} \min_{\mathbf{w}, b} & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{subject to} & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad i = 1, \dots, n. \end{array}$$

This is the key transition **Geometry of the margin**



Optimization problem

Separating hyperplane

$$\mathbf{w}^\top \mathbf{x} + b = 0$$

Summary

Point-to-boundary distance

$$d = \frac{|\mathbf{w}^\top \mathbf{x} + b|}{\|\mathbf{w}\|}$$

Geometric margin

$$\gamma_i = \frac{y_i(\mathbf{w}^\top \mathbf{x}_i + b)}{\|\mathbf{w}\|}$$

Margin boundaries

$$\mathbf{w}^\top \mathbf{x} + b = \pm 1$$

Margin width

$$M = \frac{2}{\|\mathbf{w}\|}$$

Maximum-margin classifier

Essential Formulas to Remember

Concept	Formula
Decision boundary	$\mathbf{w}^\top \mathbf{x} + b = 0$
Functional margin	$\hat{\gamma}_i = y_i(\mathbf{w}^\top \mathbf{x}_i + b)$
Geometric margin	$\gamma_i = \frac{y_i(\mathbf{w}^\top \mathbf{x}_i + b)}{\ \mathbf{w}\ }$
Point-to-hyperplane distance	$d = \frac{ \mathbf{w}^\top \mathbf{x} + b }{\ \mathbf{w}\ }$
Margin boundaries	$\mathbf{w}^\top \mathbf{x} + b = \pm 1$
Margin width	$M = \frac{2}{\ \mathbf{w}\ }$
Maximum-margin objective	$\min \frac{1}{2} \ \mathbf{w}\ ^2$

First Numerical Example: Hard-Margin SVM

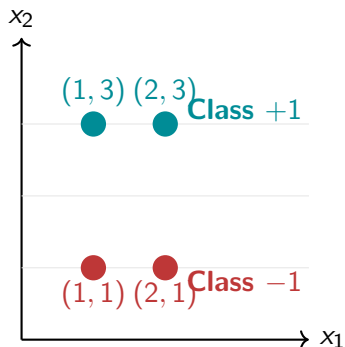
Consider the following two-class dataset:

Class + 1	Class - 1
(1, 3)	(1, 1)
(2, 3)	(2, 1)

Assume the data are linearly separable. **Find:**

- 1 possible separating boundaries,
- 2 the maximum-margin boundary,
- 3 $\mathbf{w}$ and b ,
- 4 the support vectors,
- 5 the margin,
- 6 distances from the observations to the boundary,
- 7 the class of a new observation (1.5, 2.5), and verify the SVM constraints.

Step 1: Plot the Training Observations



Observe the geometry

The positive observations lie at

$$x_2 = 3.$$

The negative observations lie at

$$x_2 = 1.$$

There is a clear empty region between the two classes.

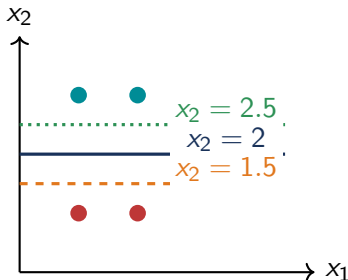
The data are linearly separable.

Step 2: Identify Possible Separating Boundaries

Any horizontal line between $x_2 = 1$ and $x_2 = 3$ separates the two classes.

For example: $x_2 = 1.5$ $x_2 = 2$ $x_2 = 2.5$

All three give zero training errors.



Key observation

Perfect separation does not give a unique boundary.

We need another criterion.

Step 3: Compare the Candidate Margins

For this simple dataset, the closest positive points are at $x_2 = 3$ and the closest negative points are at $x_2 = 1$.

Therefore:

Boundary	Distance to closest observations
$x_2 = 1.5$	0.5
$x_2 = 2.0$	1.0
$x_2 = 2.5$	0.5

Thus the total margin is:

Boundary	Total margin
$x_2 = 1.5$	1
$x_2 = 2.0$	2
$x_2 = 2.5$	1

Therefore, $x_2 = 2$
has the largest margin.

Maximum-margin decision boundary: $x_2 = 2$

Step 4: Why Is $x_2 = 2$ the Maximum-Margin Boundary?

The positive observations are at

$$x_2 = 3.$$

The negative observations are at

$$x_2 = 1.$$

The midpoint between them is

$$\frac{3 + 1}{2} = 2.$$

Therefore the maximum-margin boundary is exactly halfway between the two classes:

$$x_2 = 2$$

The distances are

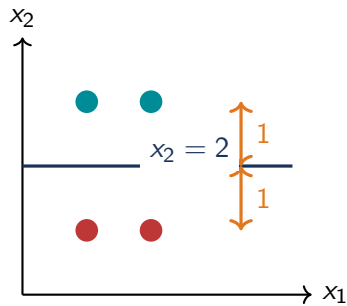
$$3 - 2 = 1 \text{ and } 2 - 1 = 1.$$

Hence

$$\text{distance to each class} = 1.$$

Therefore,

$$\text{total margin} = 1 + 1 = 2.$$



Step 5: Determine w and b

The decision boundary is

$$x_2 = 2.$$

Write it in the standard SVM form:

$$w_1x_1 + w_2x_2 + b = 0.$$

Since the boundary does not depend on x_1 ,

$$w_1 = 0.$$

Choose

$$w_2 = 1.$$

Then

$$x_2 + b = 0.$$

Since the boundary is $x_2 = 2$,

$$2 + b = 0.$$

Therefore,

$$b = -2.$$

Hence

$$\mathbf{w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad b = -2$$

and the classifier is

$$f(\mathbf{x}) = x_2 - 2.$$

Decision rule

$$\hat{y} = \text{sign}(x_2 - 2).$$

Therefore:

$$x_2 > 2 \Rightarrow +1, \quad x_2 < 2 \Rightarrow -1.$$

Step 6: Determine the Margin Boundaries

We have

$$\mathbf{w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad b = -2.$$

The canonical margin boundaries are

$$\mathbf{w}^\top \mathbf{x} + b = +1$$

and

$$\mathbf{w}^\top \mathbf{x} + b = -1.$$

Since

$$\mathbf{w}^\top \mathbf{x} + b = x_2 - 2,$$

the positive margin boundary is

$$x_2 - 2 = 1$$

so

$$x_2 = 3.$$

The negative margin boundary is

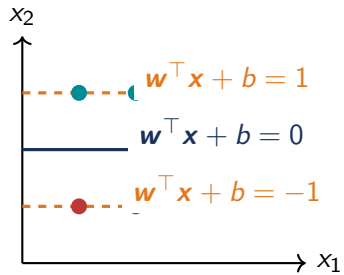
$$x_2 - 2 = -1$$

so

$$x_2 = 1.$$

Thus:

Positive margin: $x_2 = 3$,
Decision boundary: $x_2 = 2$,
Negative margin: $x_2 = 1$.



Step 7: Determine the Support Vectors

Support vectors are the observations lying on the margin boundaries.
Our margin boundaries are

$$x_2 = 3$$

and

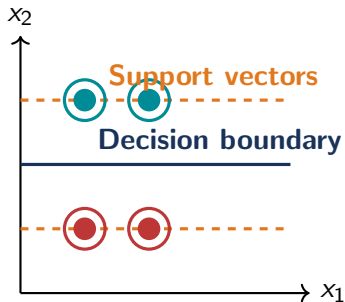
$$x_2 = 1.$$

Therefore all four training observations lie on a margin boundary:

$$(1, 3), \quad (2, 3), \quad (1, 1), \quad (2, 1).$$

Hence,

All four observations are support vectors.



Key condition

For support vectors,

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1.$$

Step 8: Verify the Support-Vector Condition

We have

$$f(\mathbf{x}) = x_2 - 2.$$

For $(1, 3)$:

$$y = +1, \quad f(\mathbf{x}) = 3 - 2 = 1.$$

Therefore,

$$yf(\mathbf{x}) = 1(1) = 1.$$

For $(2, 3)$:

$$y = +1, \quad f(\mathbf{x}) = 1.$$

Therefore,

$$yf(\mathbf{x}) = 1.$$

For (1, 1):

$$y = -1, \quad f(\mathbf{x}) = 1 - 2 = -1.$$

Therefore,

$$yf(\mathbf{x}) = (-1)(-1) = 1.$$

For (2, 1):

$$y = -1, \quad f(\mathbf{x}) = -1.$$

Therefore,

$$yf(\mathbf{x}) = 1.$$

Thus,

$y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1 \quad \text{for all four observations.}$

Step 9: Calculate the Margin

We have

$$\mathbf{w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$\|\mathbf{w}\| = \sqrt{0^2 + 1^2} = 1.$$

The total margin is

$$M = \frac{2}{\|\mathbf{w}\|}.$$

Hence,

$$M = \frac{2}{1}.$$

Therefore,

$$\boxed{M = 2}.$$

The distance from the decision boundary to either margin boundary is

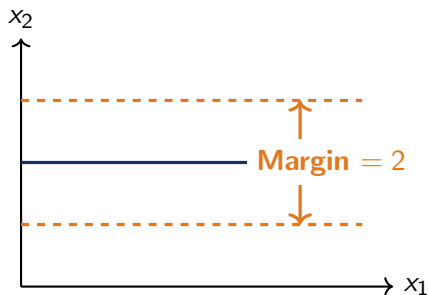
$$\frac{1}{\|\mathbf{w}\|} = 1.$$

Thus:

distance on each side = 1

and

total margin width = 2.



Step 10: Distance of Each Observation from the Boundary

The distance formula is

$$d = \frac{|\mathbf{w}^\top \mathbf{x}_i + b|}{\|\mathbf{w}\|}.$$

Here,

$$\mathbf{w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad b = -2, \quad \|\mathbf{w}\| = 1.$$

Therefore,

$$d = |x_2 - 2|.$$

Observation	y_i	$f(\mathbf{x}_i)$	Distance
(1, 3)	+1	1	1
(2, 3)	+1	1	1
(1, 1)	-1	-1	1
(2, 1)	-1	-1	1

Hence every training observation is exactly

1 unit

away from the decision boundary.

Geometric interpretation

The four observations are precisely the points that touch the margin boundaries.
Therefore they are all support vectors.

Step 11: Classify a New Observation

Suppose a new observation arrives:

$$\mathbf{x}_{\text{new}} = \begin{bmatrix} 1.5 \\ 2.5 \end{bmatrix}.$$

Our classifier is

$$\hat{y} = \text{sign}(x_2 - 2).$$

Substitute $x_2 = 2.5$:

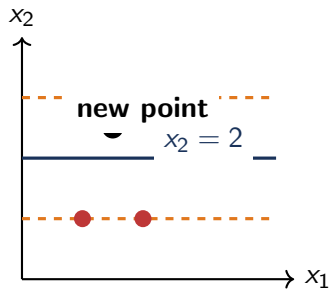
$$f(\mathbf{x}_{\text{new}}) = 2.5 - 2 = 0.5.$$

Since

$$0.5 > 0,$$

we obtain

$$\hat{y} = +1.$$



Step 12: Verify the Hard-Margin Constraints

For a hard-margin SVM, every training observation must satisfy

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1.$$

We have

$$\mathbf{w}^\top \mathbf{x} + b = x_2 - 2.$$

Therefore:

$\mathbf{x}_i$	y_i	$y_i(\mathbf{w}^\top \mathbf{x}_i + b)$	$\geq 1?$
(1, 3)	+1	(+1)(1) = 1	✓
(2, 3)	+1	(+1)(1) = 1	✓
(1, 1)	-1	(-1)(-1) = 1	✓
(2, 1)	-1	(-1)(-1) = 1	✓

Therefore all constraints are satisfied:

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 \quad \forall i.$$

And for every observation,

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1.$$

Thus all four observations lie exactly on the margin boundaries.

Complete Solution at a Glance

Quantity	Answer
Training data	$(1, 3), (2, 3), (1, 1), (2, 1)$
Maximum-margin boundary	$x_2 = 2$
Weight vector	$\mathbf{w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
Intercept	$b = -2$
Classifier	$f(\mathbf{x}) = x_2 - 2$
Positive margin boundary	$x_2 = 3$
Negative margin boundary	$x_2 = 1$
Support vectors	All four observations
Distance to either margin	1
Total margin	2
New point $(1.5, 2.5)$	+1

What Did We Learn From This Example?

Small linearly separable dataset

Several possible separating boundaries

Compare distances to the closest observations

Choose the largest margin

$$\mathbf{w}^\top \mathbf{x} + b = 0$$

The SVM idea

Separate the classes → maximize the margin → identify support vectors